

Matplotlib Tool Module

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2026-09-16

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Matplotlib Tool Module



Building Blocks of Computer-Assisted Data Analysis by Lukas Arnold, Simone Arnold, Florian Bagemihl, Matthias Baitsch, Marc Fehr, Maik Poetzsch and Sebastian Seipel. “Matplotlib Tool Module” is licensed under CC BY 4.0. The work is available at <https://github.com/bausteine-der-datenanalyse/w-python-matplotlib-en>. Excluded from the license are all logos and other content marked otherwise. 2024

Suggested Citation

Arnold, Lukas, Simone Arnold, Matthias Baitsch, Marc Fehr, Maik Poetzsch, and Sebastian Seipel. 2024. “Building Blocks of Computer-Assisted Data Analysis. Matplotlib Tool Module”. <https://github.com/bausteine-der-datenanalyse/w-python-matplotlib-en>.

BibTeX Template

```
@misc{BCD-w-python-matplotlib-2024,  
  title={Building Blocks of Computer-Assisted Data Analysis. Matplotlib Tool Module},  
  author={Arnold, Lukas and Arnold, Simone and Baitsch, Matthias and Fehr, Marc and Poetzsch, Maik and Seipel, Sebastian},  
  year={2024},  
  url={https://github.com/bausteine-der-datenanalyse/w-python-matplotlib-en}}
```

Intro

Prerequisites

- Python basics
- Importing additional packages

Packages and Datasets Used

- matplotlib

Estimated Time

Estimated time: 1h

Learning Objectives

- Introduction: how to visualize data in Python
- Customizing plots
- Do's & Don'ts for scientific plots

1 Introduction to Matplotlib

Matplotlib is one of the most well-known libraries for data visualization in Python. It allows the creation of static, animated, and interactive plots with high flexibility.

1.1 Why Matplotlib?

- **Broad Support:** Works well with NumPy, Pandas, and SciPy.
- **High Customizability:** Full control over plots.
- **Integration with Jupyter Notebooks:** Ideal for interactive data analysis.
- **Compatibility:** Supports various output formats (PNG, SVG, PDF, etc.).

1.2 Alternatives to Matplotlib

While Matplotlib is powerful, there are alternatives that may be better suited for specific purposes: - **Seaborn:** Built on top of Matplotlib, simplifies statistical visualizations. - **Plotly:** Creates interactive plots, great for dashboards. - **Bokeh:** Ideal for web applications with interactive visualizations.

1.3 First Example: Plotting a Simple Line

```
import matplotlib.pyplot as plt
import numpy as np

# Example data
t = np.linspace(0, 10, 100)
y = np.sin(t)

# Create the plot
plt.plot(t, y, label='sin(t)')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
```

```
plt.title('Simple Line Plot')  
plt.legend()  
plt.show()
```

This simple example demonstrates how to visualize a **sine curve** using Matplotlib.

1.4 Next Steps

In the next chapter, we will explore the different types of plots that Matplotlib offers.

2 Plot Types in Matplotlib

Matplotlib offers a variety of plot types suitable for different purposes. In this chapter, we introduce the most important plot types and explain their use cases.

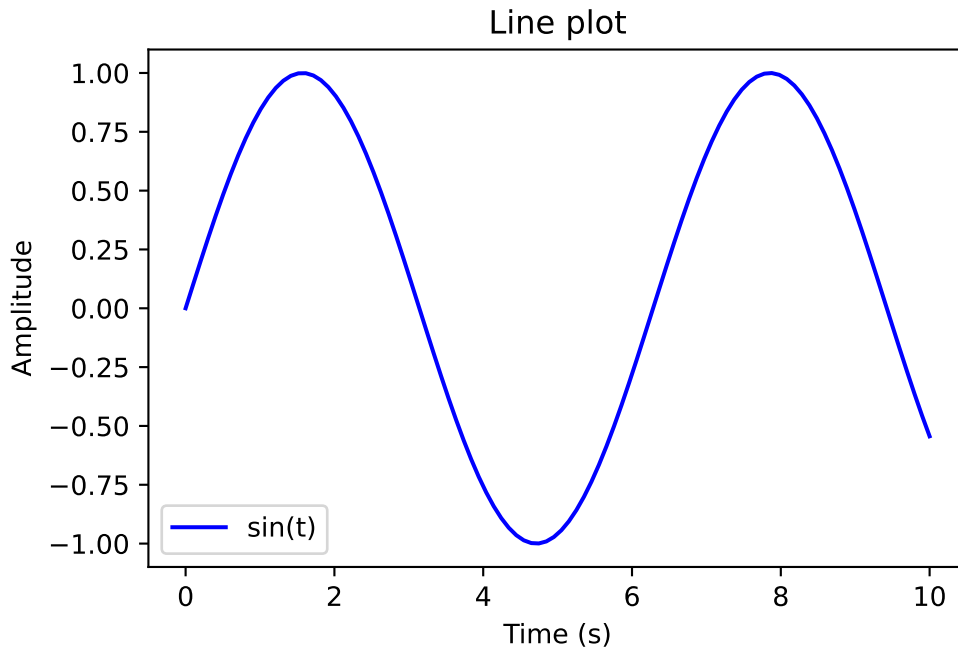
2.1 1. Line plots (`plt.plot()`)

Line plots are excellent for visualizing trends over time.

```
import matplotlib.pyplot as plt
import numpy as np

t = np.linspace(0, 10, 100)
y = np.sin(t)

plt.plot(t, y, label='sin(t)', color='b')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.title('Line plot')
plt.legend()
plt.show()
```

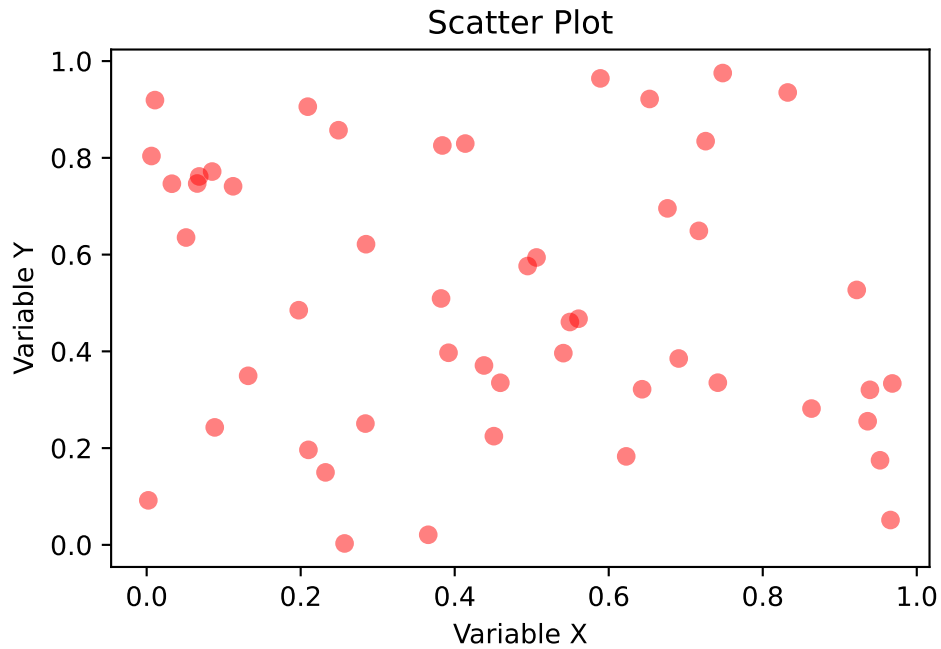


2.2 2. Scatter Plots (plt.scatter())

Scatter plots are used to show relationships between two variables.

```
x = np.random.rand(50)
y = np.random.rand(50)

plt.scatter(x, y, color='r', alpha=0.5)
plt.xlabel('Variable X')
plt.ylabel('Variable Y')
plt.title('Scatter Plot')
plt.show()
```

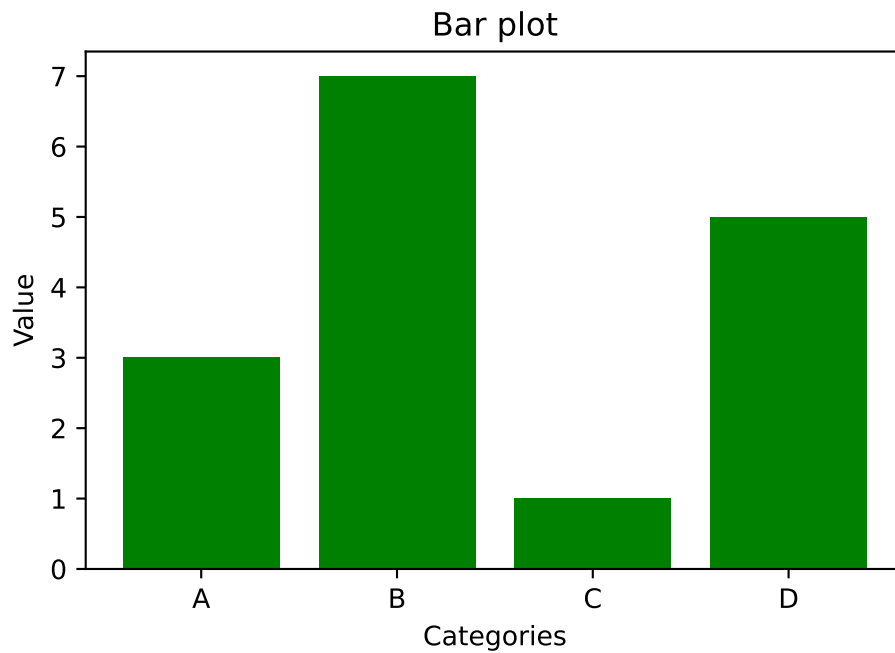


2.3 3. Bar plots (plt.bar())

Bar plots are suitable for representing categorical data.

```
categories = ['A', 'B', 'C', 'D']
values = [3, 7, 1, 5]

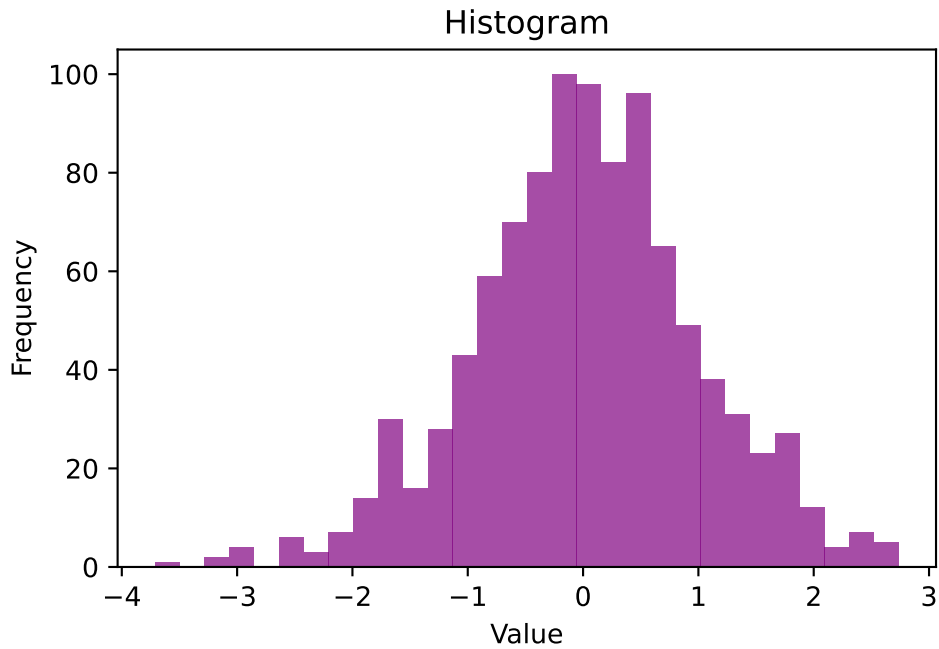
plt.bar(categories, values, color='g')
plt.xlabel('Categories')
plt.ylabel('Value')
plt.title('Bar plot')
plt.show()
```



2.4 4. Histograms (plt.hist())

Histograms show the distribution of numerical data.

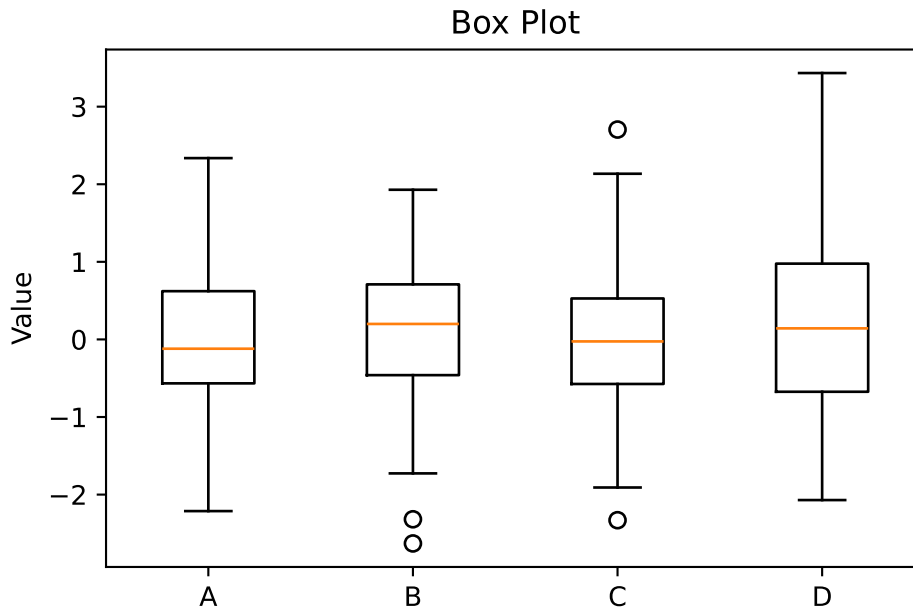
```
data = np.random.randn(1000)
plt.hist(data, bins=30, color='purple', alpha=0.7)
plt.xlabel('Value')
plt.ylabel('Frequency')
plt.title('Histogram')
plt.show()
```



2.5 5. Box Plots (plt.boxplot())

Box plots help visualize outliers and data distribution.

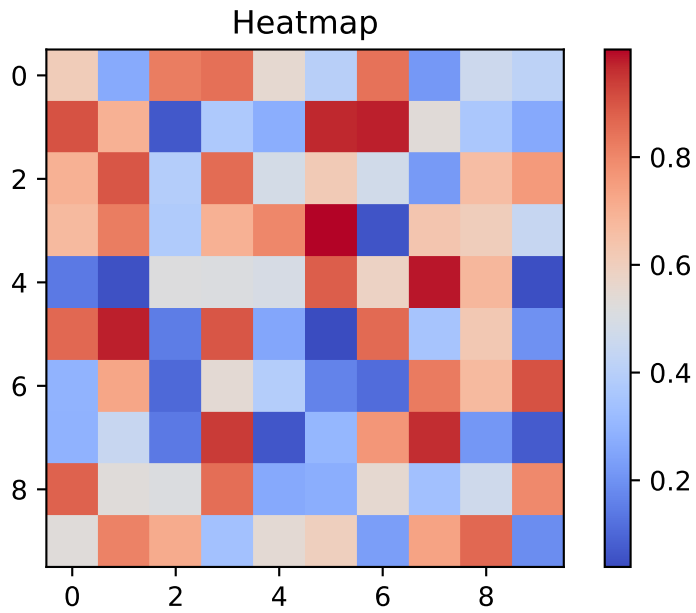
```
data = [np.random.randn(100) for _ in range(4)]
plt.boxplot(data, tick_labels=['A', 'B', 'C', 'D'])
plt.ylabel('Value')
plt.title('Box Plot')
plt.show()
```



2.6 6. Heatmaps (plt.imshow())

Heatmaps are useful for displaying 2D data.

```
data = np.random.rand(10, 10)
plt.imshow(data, cmap='coolwarm', interpolation='nearest')
plt.colorbar()
plt.title('Heatmap')
plt.show()
```



2.7 Conclusion

The choice of plot type depends on the nature of the data and the intended representation. In the next chapter, we will explore customizing and adapting plots.

3 Customization and Adapting of Plots in Matplotlib

A well-designed plot improves the readability and comprehension of the data presented. In this chapter, we will explore various ways to customize and adapt plots using Matplotlib.

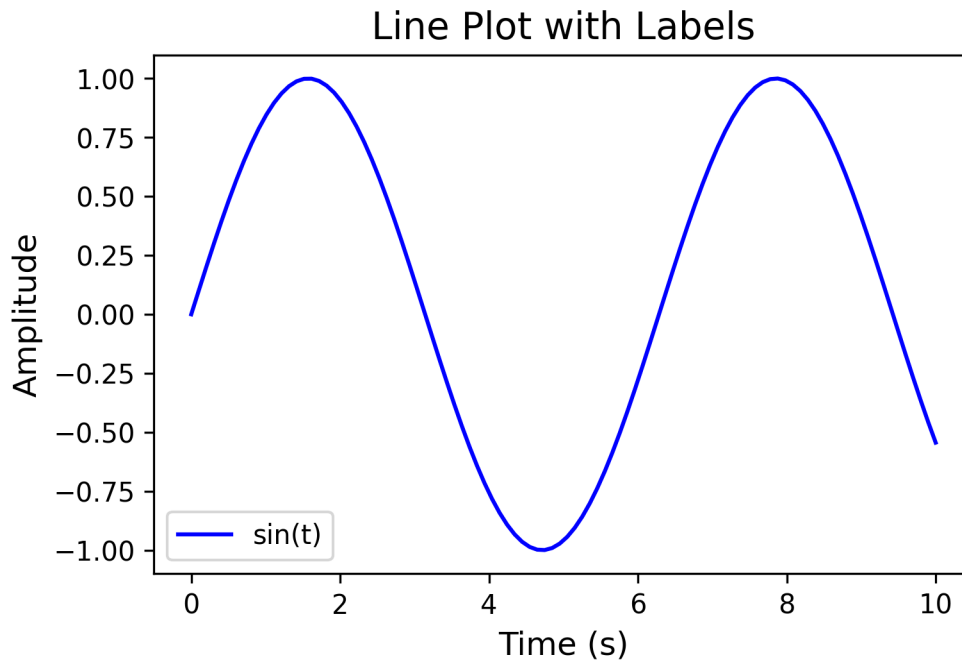
3.1 1. Axis Labels and Plot Title

Clear axis and plot titles are essential for understanding a plot.

```
import matplotlib.pyplot as plt
import numpy as np

t = np.linspace(0, 10, 100)
y = np.sin(t)

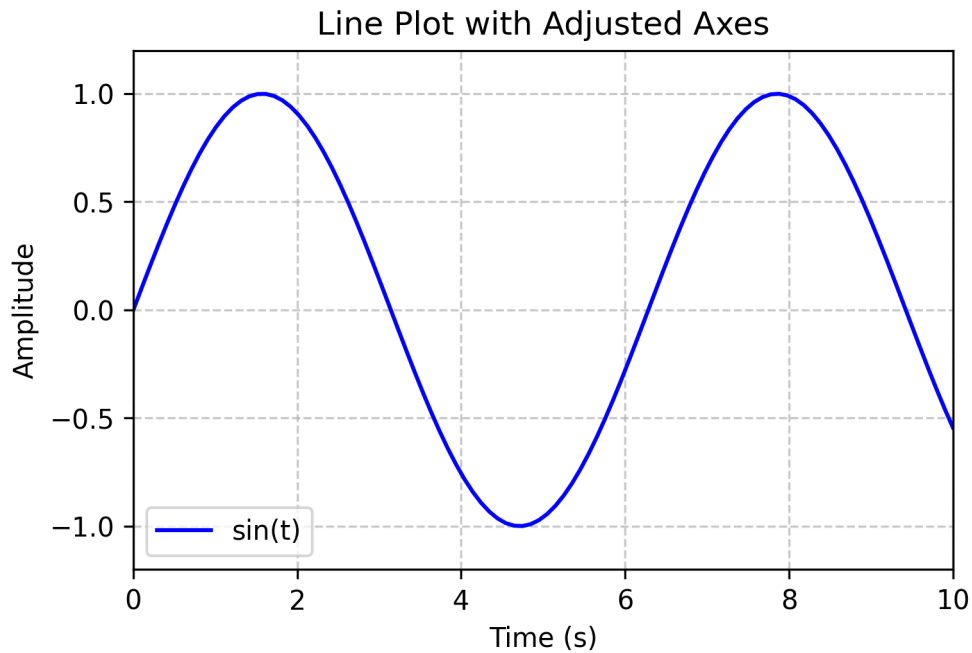
plt.plot(t, y, label='sin(t)', color='b')
plt.xlabel('Time (s)', fontsize=12)
plt.ylabel('Amplitude', fontsize=12)
plt.title('Line Plot with Labels', fontsize=14)
plt.legend()
plt.show()
```



3.2 2. Adjusting Axes

The scaling of the axes should be chosen appropriately to best represent the data.

```
plt.plot(t, y, label='sin(t)', color='b')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.xlim(0, 10)
plt.ylim(-1.2, 1.2)
plt.grid(True, linestyle='--', alpha=0.7)
plt.title('Line Plot with Adjusted Axes')
plt.legend()
plt.show()
```



3.3 3. Colors and Line Styles

Colors and line styles help to highlight important information in the plot.

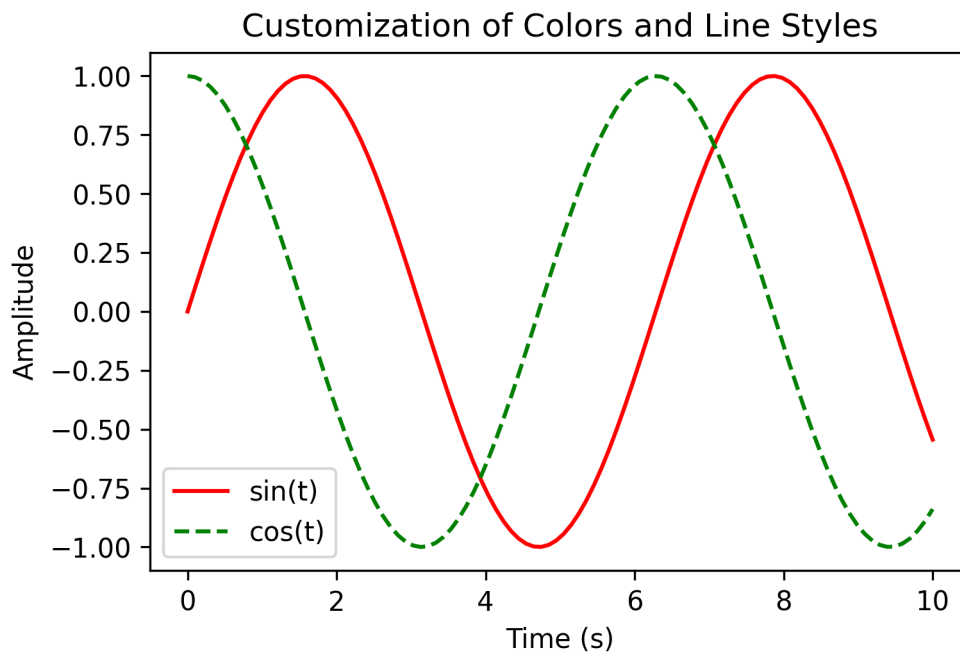
3.3.1 Common Colors (Default Colors in Matplotlib)

Color	Code	Description
Blue	'b'	blue
Green	'g'	green
Red	'r'	red
Cyan	'c'	cyan
Magenta	'm'	magenta
Yellow	'y'	yellow
Black	'k'	black
White	'w'	white

3.3.2 Common Line Styles

Line Style	Code	Description
Solid	'-'	default line
Dashed	'--'	long dashes
Dotted	'.'	only dots
Dash-dot	'-.'	alternating dash-dot

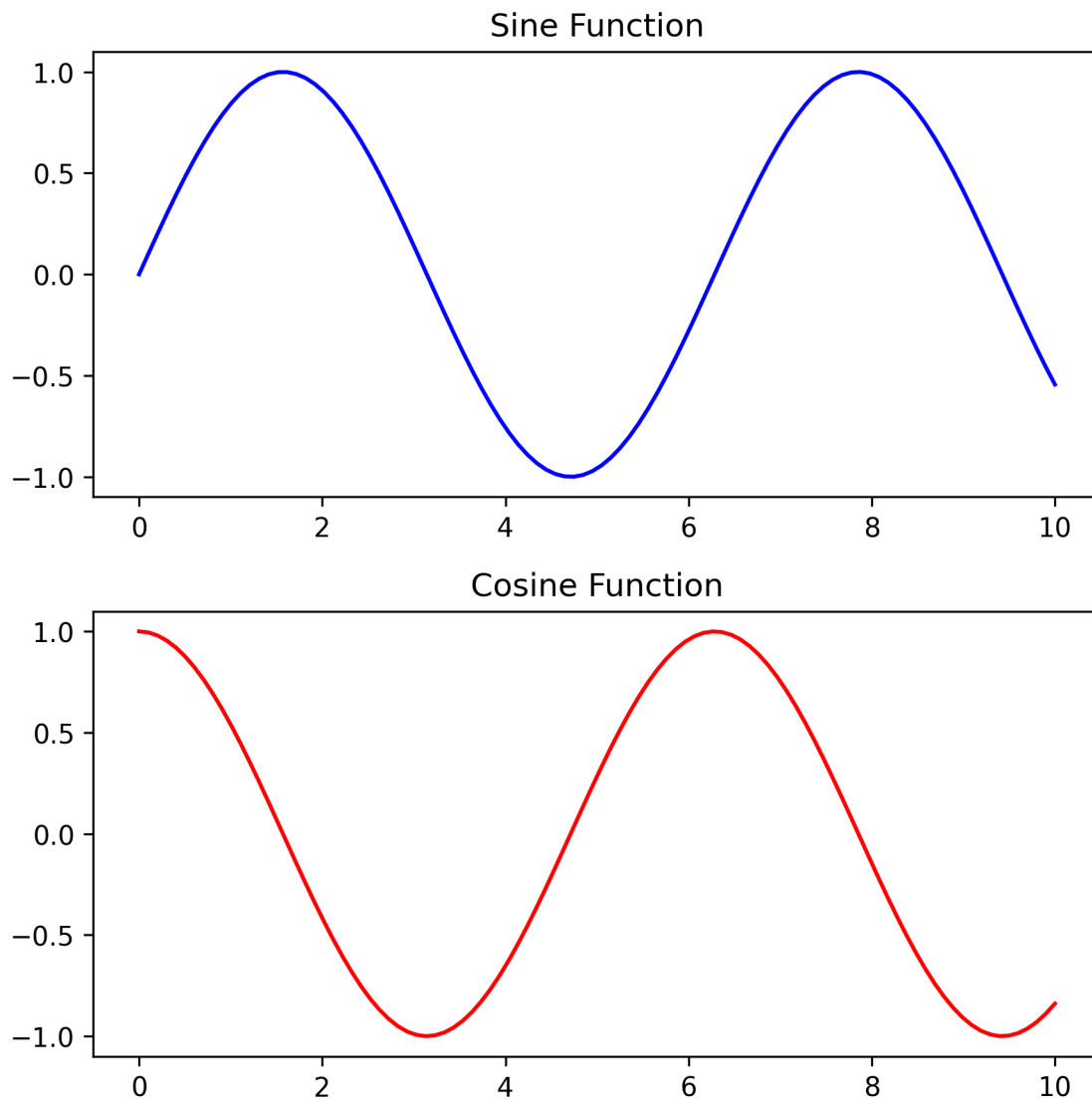
```
plt.plot(t, np.sin(t), linestyle='-', color='r', label='sin(t)')
plt.plot(t, np.cos(t), linestyle='--', color='g', label='cos(t)')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.title('Customization of Colors and Line Styles')
plt.legend()
plt.show()
```



3.4 4. Multiple Plots with Subplots

Sometimes it's useful to display multiple plots in a single figure.

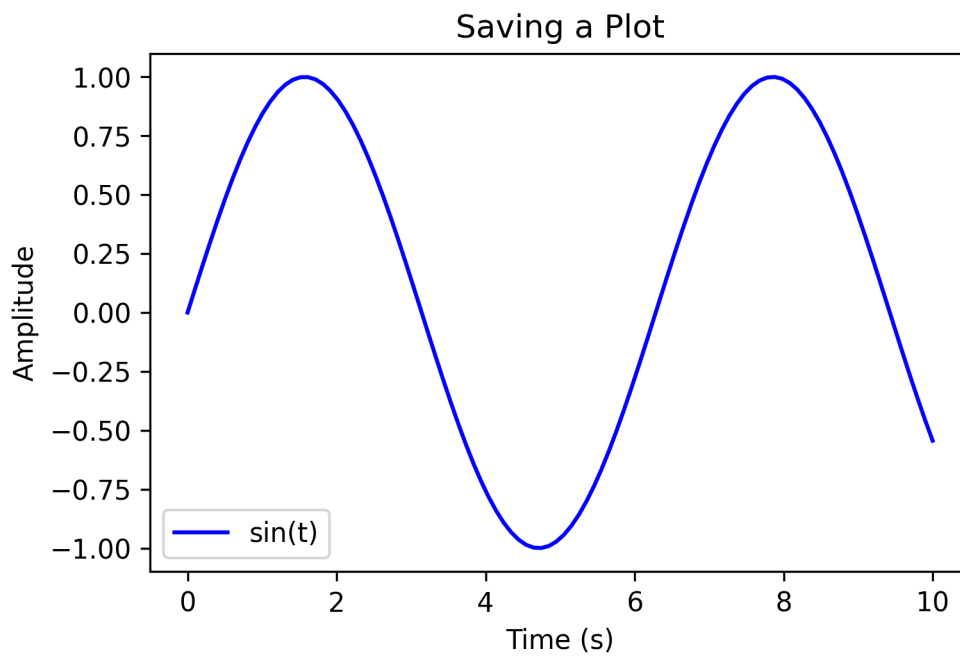
```
fig, axs = plt.subplots(2, 1, figsize=(6, 6))
axs[0].plot(t, np.sin(t), color='b')
axs[0].set_title('Sine Function')
axs[1].plot(t, np.cos(t), color='r')
axs[1].set_title('Cosine Function')
plt.tight_layout()
plt.show()
```



3.5 5. Saving Plots

Plots can be saved in various formats.

```
plt.plot(t, y, label='sin(t)', color='b')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.title('Saving a Plot')
plt.legend()
plt.savefig('my_plot.png', dpi=300)
plt.show()
```



3.6 Conclusion

With careful customization, scientific plots can be significantly improved. In the next chapter, we will explore advanced techniques such as logarithmic scales and annotations.

4 Advanced Techniques in Matplotlib

In this chapter, we explore some advanced features of Matplotlib that are especially useful for scientific data visualization.

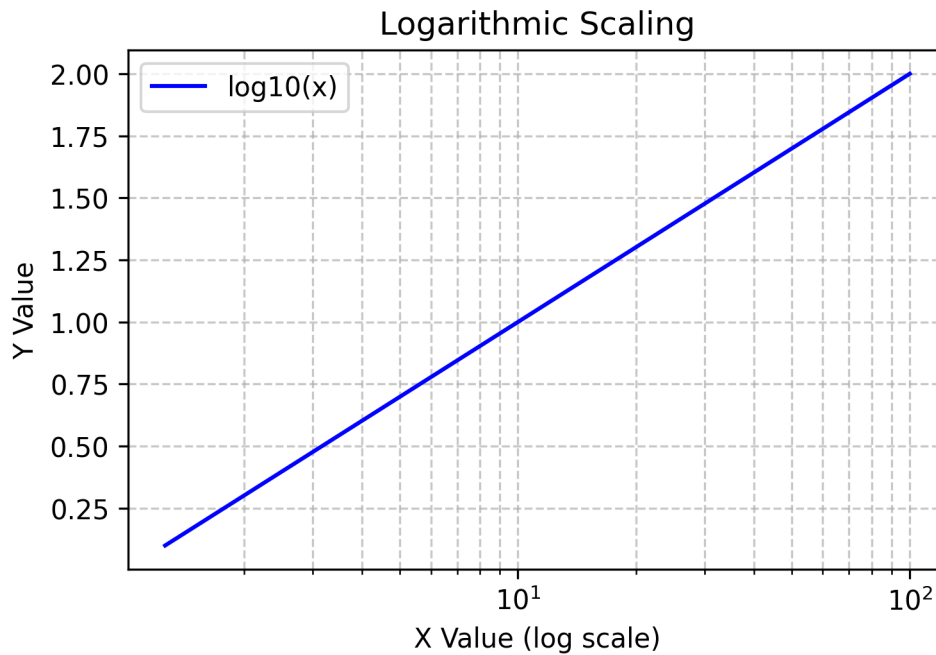
4.1 1. Logarithmic Scales

Logarithmic scales are often used when values span several orders of magnitude.

```
import matplotlib.pyplot as plt
import numpy as np

x = np.logspace(0.1, 2, 100)
y = np.log10(x)

plt.plot(x, y, label='log10(x)', color='b')
plt.xscale('log')
plt.xlabel('X Value (log scale)')
plt.ylabel('Y Value')
plt.title('Logarithmic Scaling')
plt.legend()
plt.grid(True, which='both', linestyle='--', alpha=0.7)
plt.show()
```



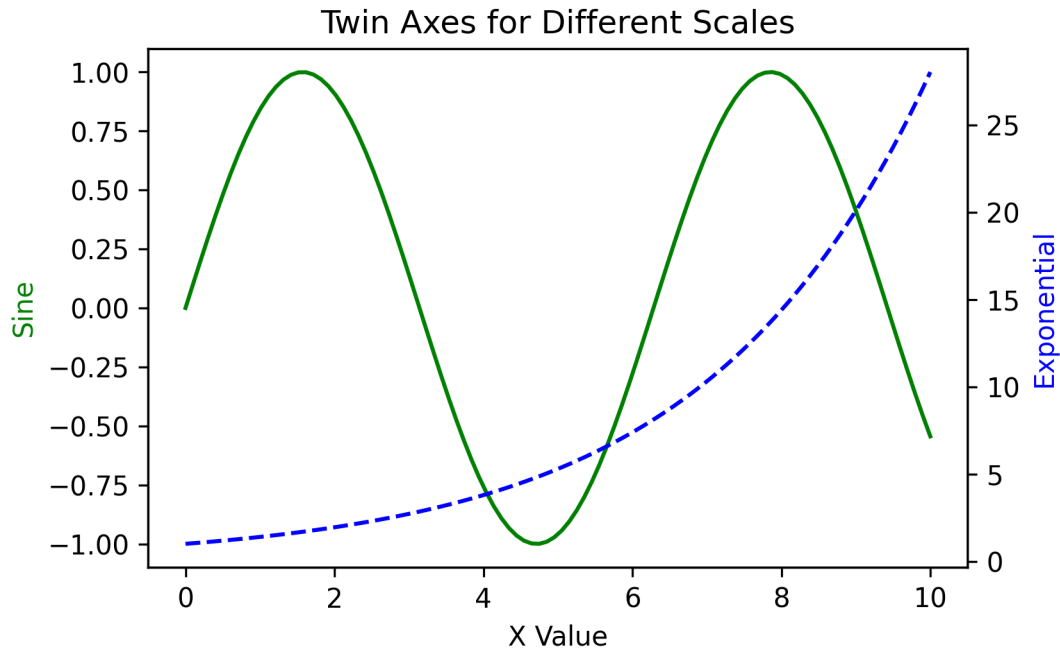
4.2 2. Twin Axes for Different Scales

Sometimes you may want to display two different y-axes in one plot.

```
x = np.linspace(0, 10, 100)
y1 = np.sin(x)
y2 = np.exp(x / 3)

fig, ax1 = plt.subplots()
ax2 = ax1.twinx()
ax1.plot(x, y1, 'g-', label='sin(x)')
ax2.plot(x, y2, 'b--', label='exp(x/3)')

ax1.set_xlabel('X Value')
ax1.set_ylabel('Sine', color='g')
ax2.set_ylabel('Exponential', color='b')
ax1.set_title('Twin Axes for Different Scales')
plt.show()
```

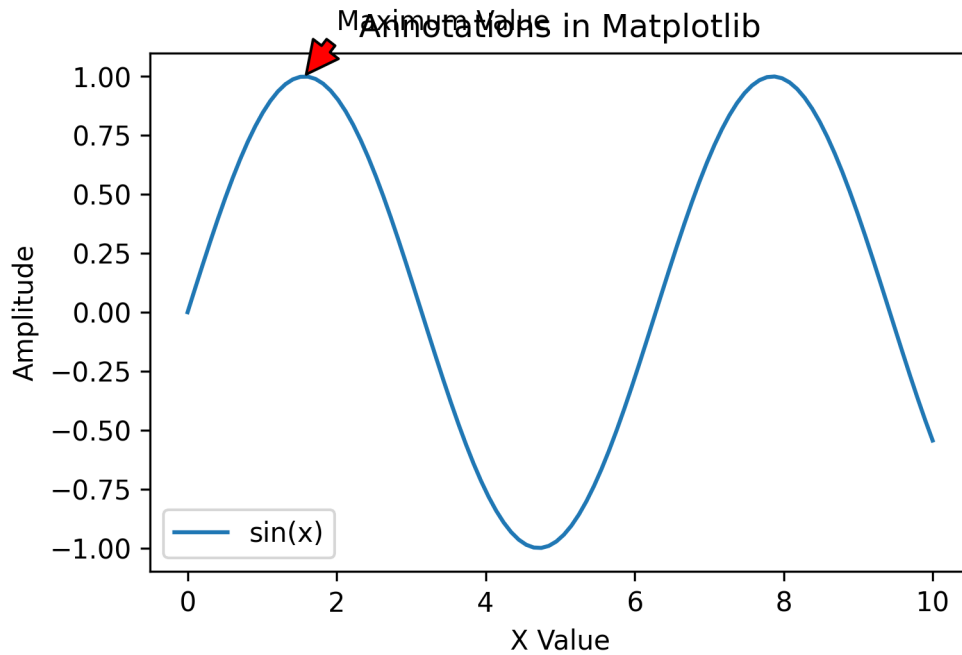


4.3 3. Annotations in Plots

Important points or values in a plot can be highlighted using annotations.

```
x = np.linspace(0, 10, 100)
y = np.sin(x)

plt.plot(x, y, label='sin(x)')
plt.xlabel('X Value')
plt.ylabel('Amplitude')
plt.title('Annotations in Matplotlib')
plt.annotate('Maximum Value', xy=(np.pi/2, 1), xytext=(2, 1.2),
            arrowprops=dict(facecolor='red', shrink=0.05))
plt.legend()
plt.show()
```



4.4 Conclusion

These advanced features help make scientific plots more informative. In the next chapter, we will look at best practices and common mistakes in scientific visualization.

5 Best Practices in Matplotlib: Common Mistakes and Improvements

In this chapter, we demonstrate a poor and an improved example for each common issue encountered when plotting with Matplotlib.

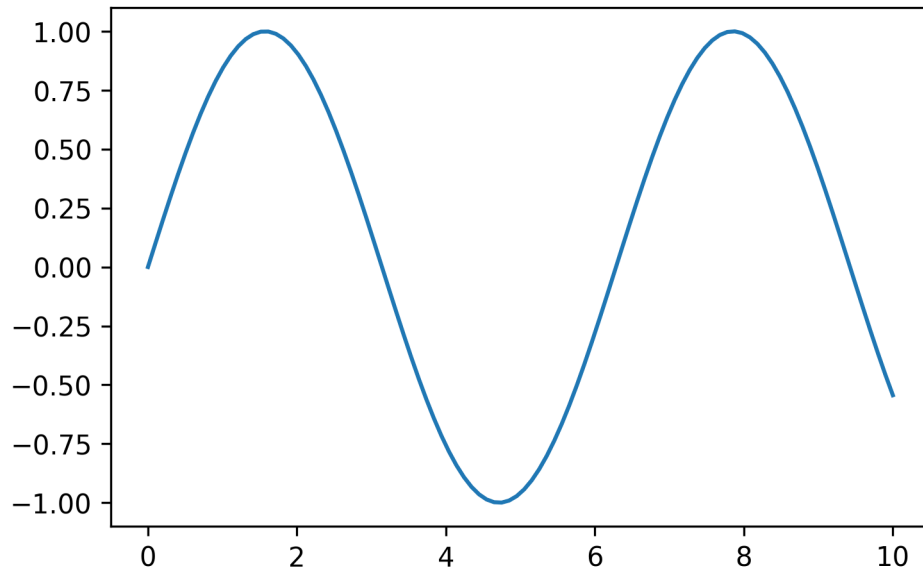
5.1 1. Missing Labels

5.1.1 Bad Example

```
import matplotlib.pyplot as plt
import numpy as np

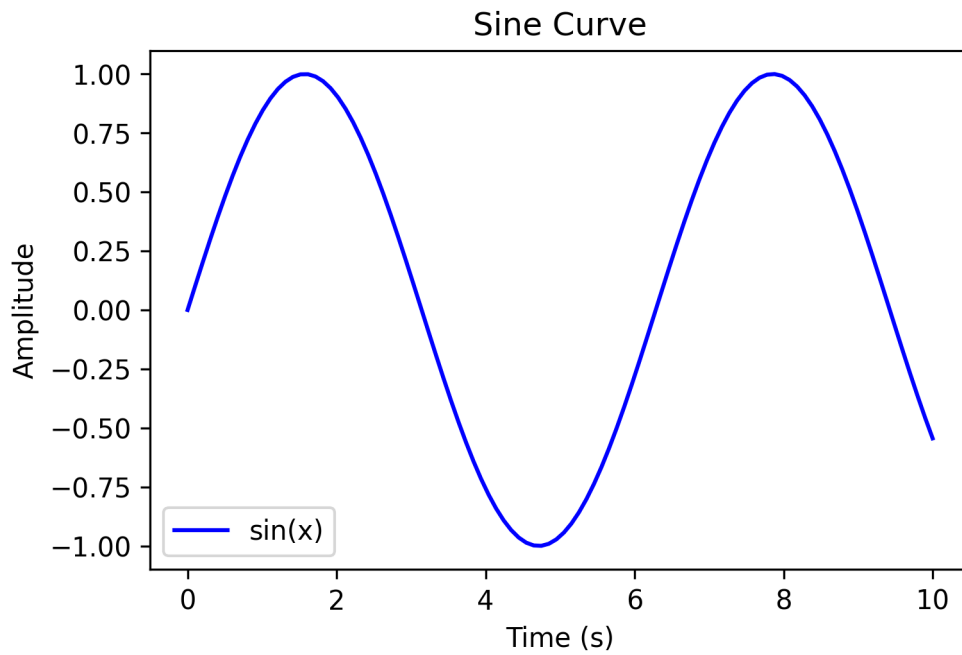
x = np.linspace(0, 10, 100)
y = np.sin(x)

plt.plot(x, y)
plt.show()
```



5.1.2 Improved Example

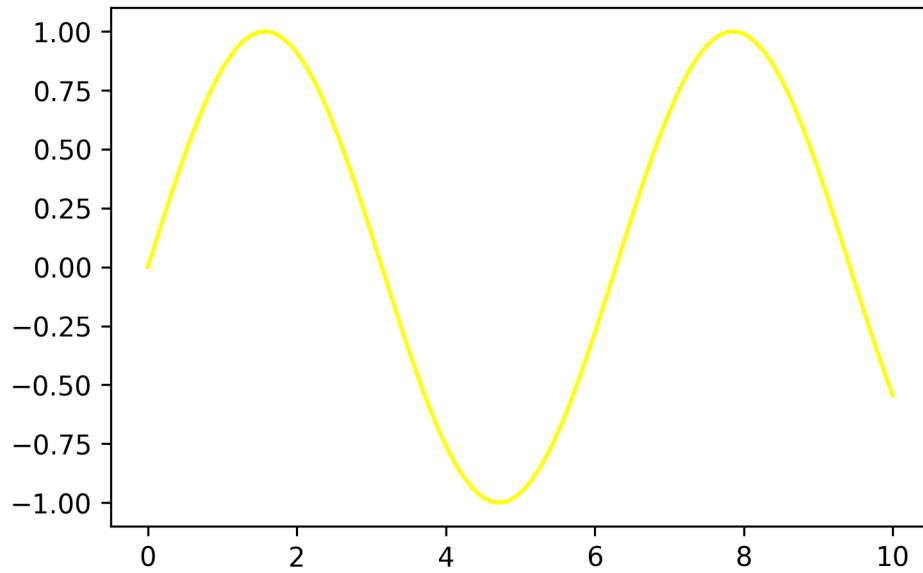
```
plt.plot(x, y, label='sin(x)', color='b')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.title('Sine Curve')
plt.legend()
plt.show()
```



5.2 2. Poor Color Choice

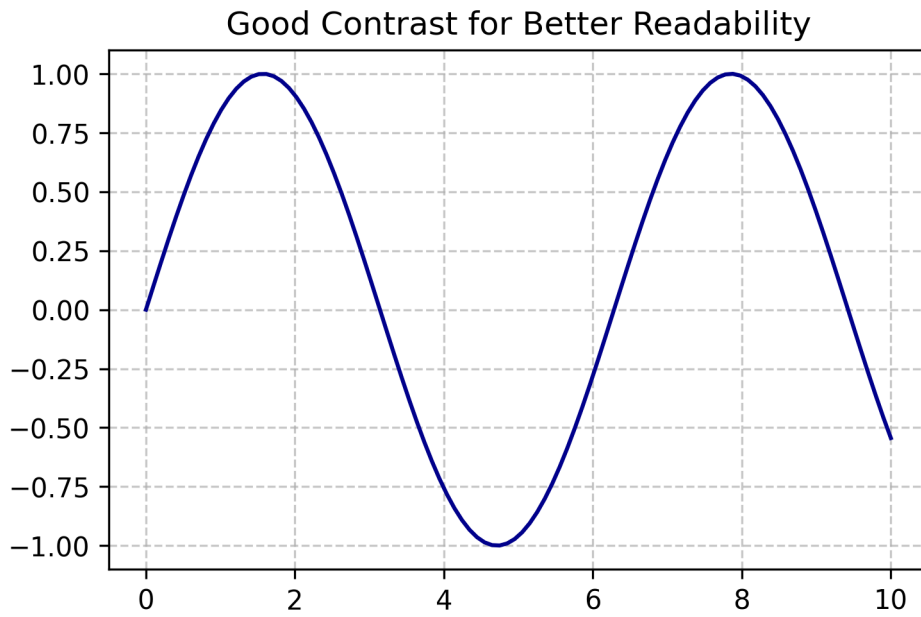
5.2.1 Bad Example

```
plt.plot(x, y, color='yellow')  
plt.show()
```



5.2.2 Improved Example

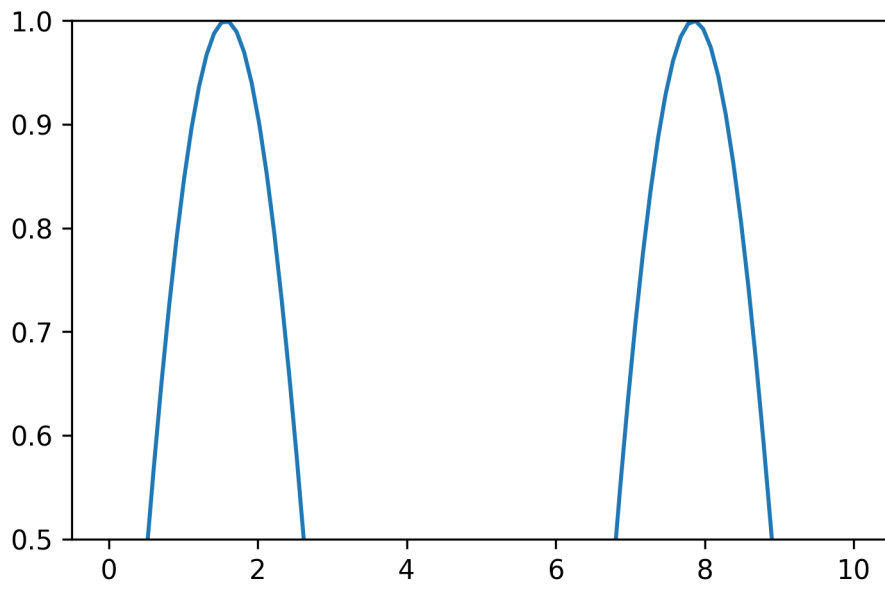
```
plt.plot(x, y, color='darkblue')
plt.grid(True, linestyle='--', alpha=0.7)
plt.title('Good Contrast for Better Readability')
plt.show()
```



5.3 3. Unreasonable Axis Scaling

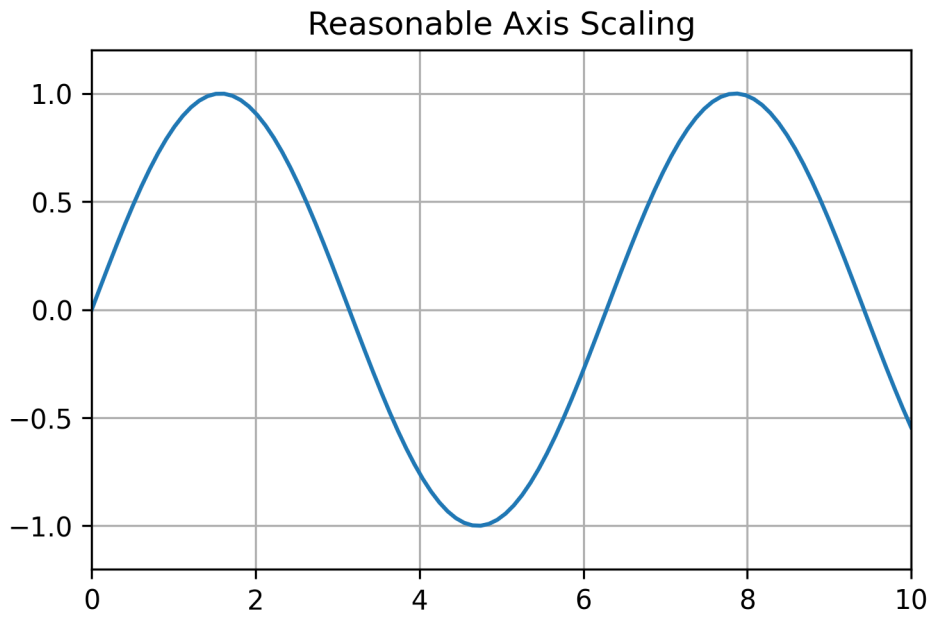
5.3.1 Bad Example

```
plt.plot(x, y)
plt.ylim(0.5, 1)
plt.show()
```



5.3.2 Improved Example

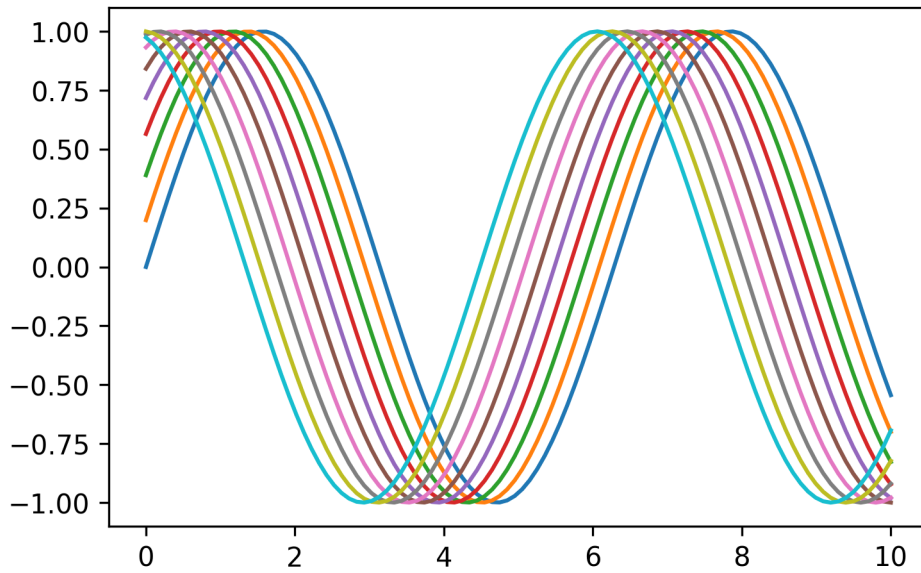
```
plt.plot(x, y)
plt.ylim(-1.2, 1.2)
plt.xlim(0, 10)
plt.grid(True)
plt.title('Reasonable Axis Scaling')
plt.show()
```



5.4 4. Overcrowded with Too Many Lines

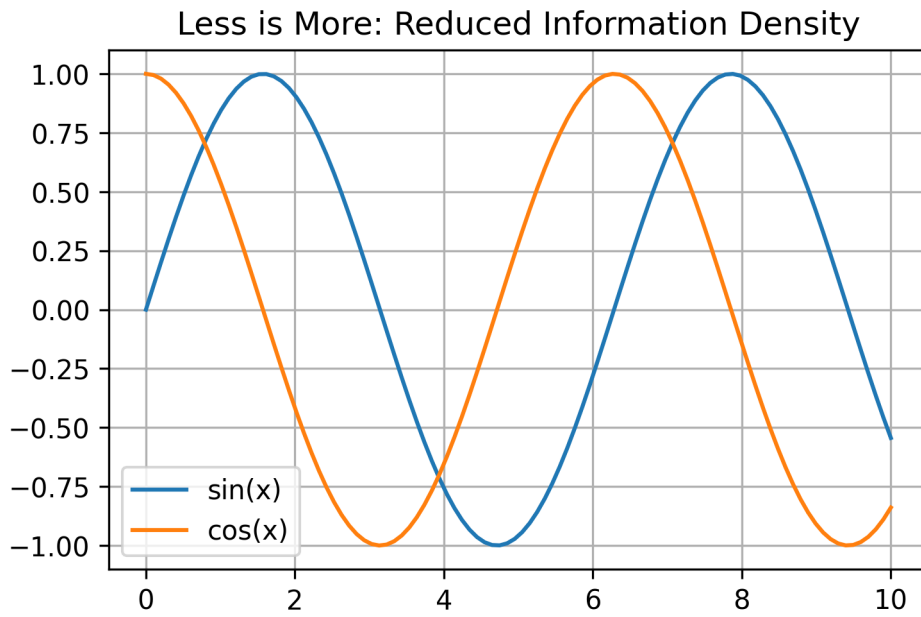
5.4.1 Bad Example

```
for i in range(10):  
    plt.plot(x, np.sin(x + i * 0.2))  
plt.show()
```



5.4.2 Improved Example

```
plt.plot(x, np.sin(x), label='sin(x)')
plt.plot(x, np.cos(x), label='cos(x)')
plt.legend()
plt.title('Less is More: Reduced Information Density')
plt.grid(True)
plt.show()
```



5.5 Conclusion

Good plots are characterized by clear labels, good readability, and a reasonable amount of information.